SPEA-E 340: Environmental Economics and Finance

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Course Description

This course explores the economic principles behind environmental problems and the policies designed to address them. We examine the foundations of markets and the ways they fail in the presence of externalities, public goods, and common-pool resources. Students learn methods for valuing environmental benefits — including travel cost, hedonic pricing, and stated preference techniques — and evaluate policy instruments such as command-and-control regulation, environmental taxes, tradable permits, and subsidies. We also study topics in environmental finance, including carbon markets and green investments.

Learning Outcomes

- Explain core theories of environmental economics, including market efficiency and market failures.
- Apply economic valuation methods, including travel cost, hedonic pricing, and stated preference techniques.
- Evaluate environmental policy instruments designed to address specific market failures.
- Compare and interpret environmental policies across time and across the U.S. and other countries.
- Assess and defend policy recommendations using efficiency, equity, and unintended-consequences analyses.
- Communicate policy insights effectively through written work and presentations.

Texts

Required: Keohane, N. O., & Olmstead, S. M. (2016). *Markets and the Environment* (2nd ed.). Island Press. Either edition is acceptable; referred to below as K&O.

Optional: Jaeger, W. K. (2005). *Environmental Economics for Tree Huggers and Other Skeptics*. Island Press.

Journal articles, reports, podcasts, and videos supplement the textbook, particularly for environmental justice, informatics, ESG finance, and recent U.S. legislation.

Assessment

Component	Weight
In-class quizzes (≈19 unannounced; lowest 3 dropped)	20%
Assignments (10; lowest dropped)	20%
Midterm exam (in class, Modules 1–2)	30%
Final exam (cumulative)	30%

Quizzes are short (≈5 minutes, three questions), closed-note, and draw on current and earlier material. They cannot be made up.

Assignments lose 1% per hour after the deadline; no credit after 48 hours.

Exams are closed-note and on paper. There are no make-up exams; a missed midterm's weight rolls into the final (60%).

A+	A	A–	B+	B	B–	C+	C	C–	D+	D	D–	F
97–100	93–97	87–92	82–86	75–81	69–74	62–68	55–61	50–54	40–49	30–39	20–29	<20

Generative AI Policy

AI may be used only to locate sources (which students must read and cite themselves), to check understanding of a concept already worked through, and for spelling and grammar. All other use — including generating answers, graphs, or any portion of submitted work — is prohibited. No AI use of any kind is permitted on exams or quizzes.

Topics and Readings

Lec	Topic	K&O Reading
Module 1 — Introduction to Environmental Economics		
1	Introduction to Environmental Economics	Ch. 1
2	Market Failures	Ch. 5
3	Property Rights (Coase)	Ch. 6, 8
4	Standards, Taxes, Permits & Subsidies (Pigou)	Ch. 8
5	Collective Action (Ostrom)	Ch. 5 + supplements
Module 2 — Measuring Environmental Benefits and Costs		
6	Econometrics	Ch. 3 + supplements
7	Environmental Informatics	Supplements
8	Discounting and Cost–Benefit Analysis	Ch. 2
9	Travel Cost Method	Ch. 3
10	Contingent Valuation	Ch. 3
11	Stated Preference & Defensive Behavior	Ch. 3 + supplements
12	Hedonics & Real Estate Markets	Ch. 3
13	Carbon Accounting & Offset Markets	Ch. 8–11
14	Climate Risk & Financial Instruments (Green & ESG)	Ch. 11 + supplements
15	Natural Capital Accounting	Ch. 11
Midterm		
16	Midterm Review	Ch. 1–11 (selective)
17	Midterm Exam (Modules 1–2)	
Module 3 — Environmental Policy and Regulation		
18	Economics of Environmental Regulation	Ch. 2, 8, 9
19	Clean Air Act	Ch. 10 + supplements
20	Clean Water Act	Ch. 10 + supplements
21	Safe Drinking Water Act	Supplements
22	Hazardous Wastes (CERCLA & RCRA)	Ch. 2 + supplements
23	Regulating Cars	Ch. 9–10
Module 4 — Topics in Environmental Economics & Policy		
24	Forests & Fisheries	Ch. 7
25	Environmental Justice in Regulation I (U.S.)	Supplements
26	Environmental Justice in Regulation II (Global)	Supplements
27	Market-Based Environmental Regulation in Practice	Ch. 9–10
28	What's Next? IRA, BIL & West Virginia v. EPA	Supplements
29	Final Exam Review	Cumulative

Topics may vary slightly by semester. This is an abbreviated public version of the syllabus; enrolled students should refer to the full syllabus on Canvas for dates, course policies and university resources.